

Burst-Controlled Phantom-Queue Policer in eBPF

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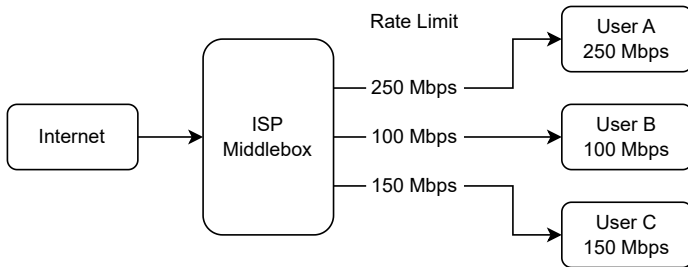
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Internet and Networked Systems Project
supervised by Jonas Köppeler

Rate Limiting

use cases

- usage quotas
- fairness
- prioritisation



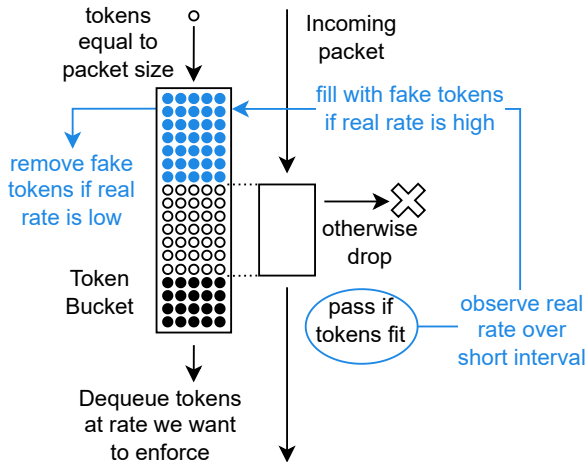
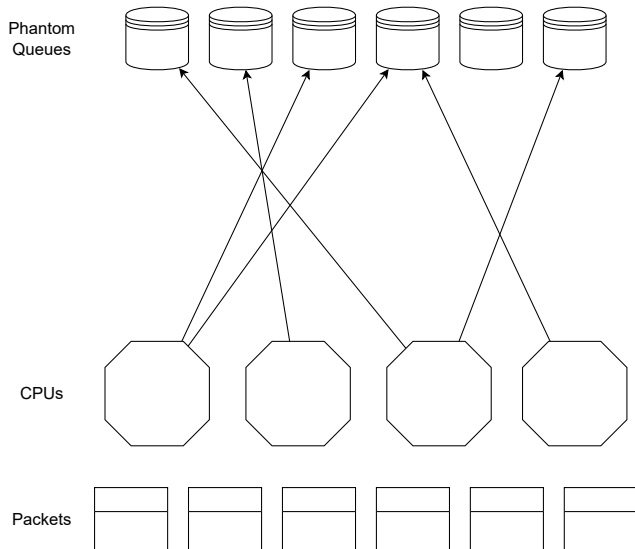


Figure 1: Burst-controlled phantom-queue policer (BC-PQP) [1]

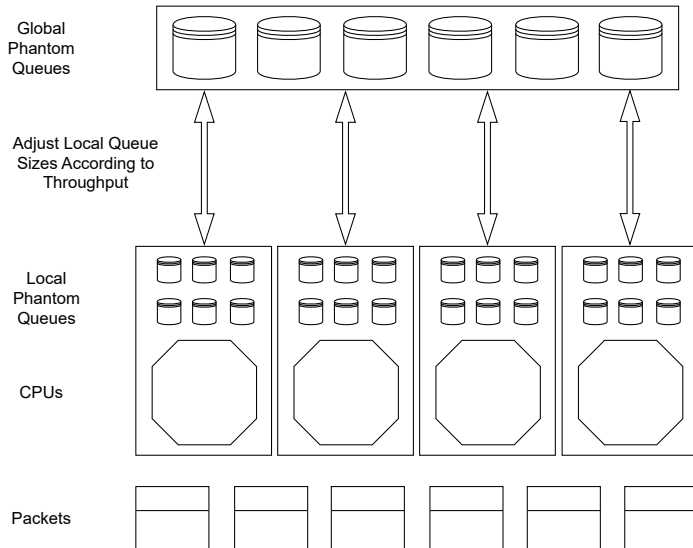
BC-PQP implementation in eBPF

- portability
- multiple cores
- integration with Linux networking stack
- application safety

Implementations: Simple



Implementations: Sharded



DEMO

Progress: Bursty Traffic Results

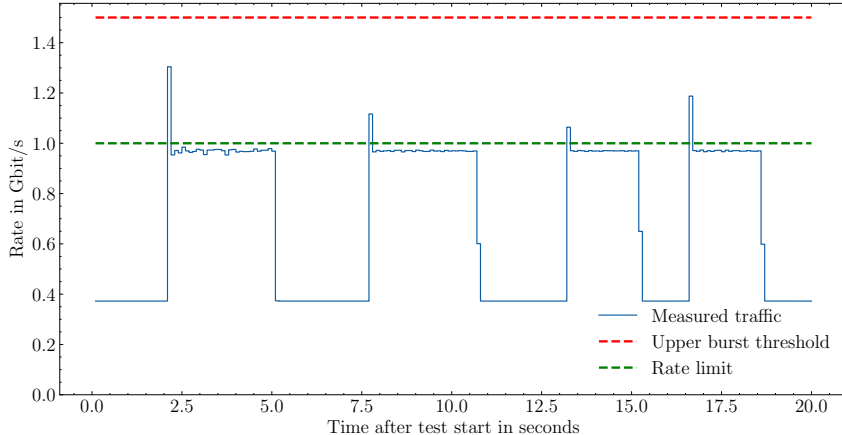


Figure 2: Simple version: Throughput for bursty UDP traffic - base sending rate of 400 Mbit/s with bursts of around 6 Gbit/s

Progress: TCP Results

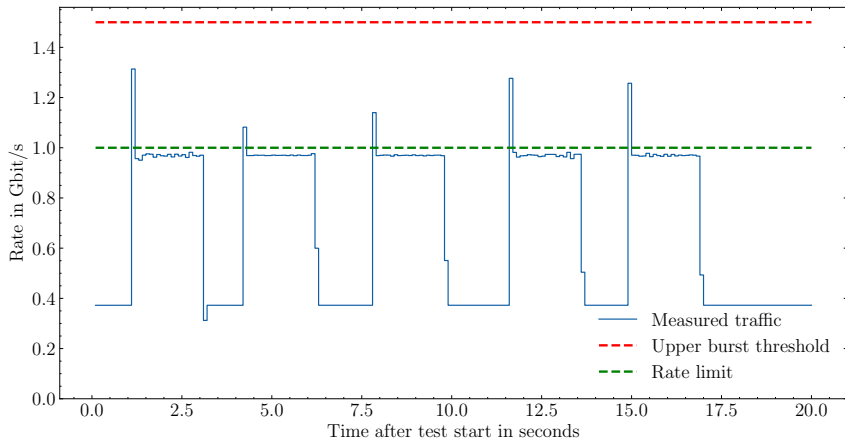


Figure 3: Simple version: Throughput for bursty TCP traffic (CUBIC) - base sending rate of 400 Mbit/s with bursts of around 6 Gbit/s

Progress: Bursty Traffic Results

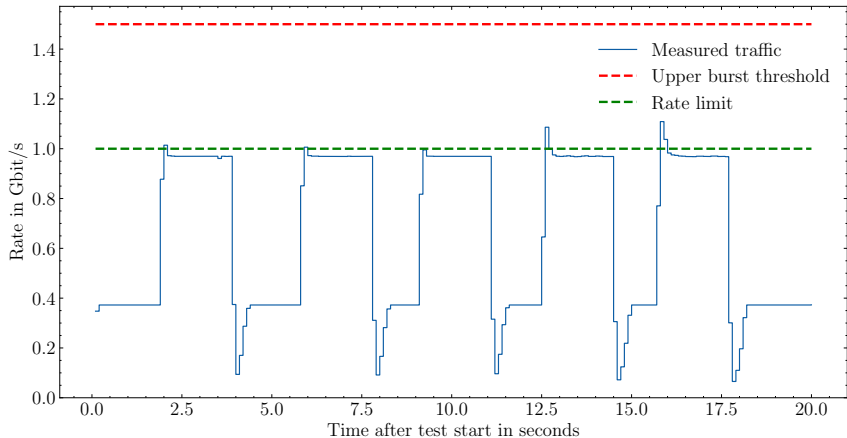


Figure 4: Sharded version: Throughput for bursty UDP traffic - base sending rate of 400 Mbit/s with bursts of around 6 Gbit/s

Real-World Experiments: Setup

Two servers

- sender-receiver: Generates traffic and measures throughput
- rate limiter: BC-PQP program loaded
- identical hardware: 20-core CPU (40 with hyperthreading), 25 GbE NICs

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What do we want to test?

- rate enforcement: Do we properly enforce the rate limit for all rates?
- scaling: What is the throughput for different numbers of CPU cores?

Real-World Experiments: Rate Enforcement Results

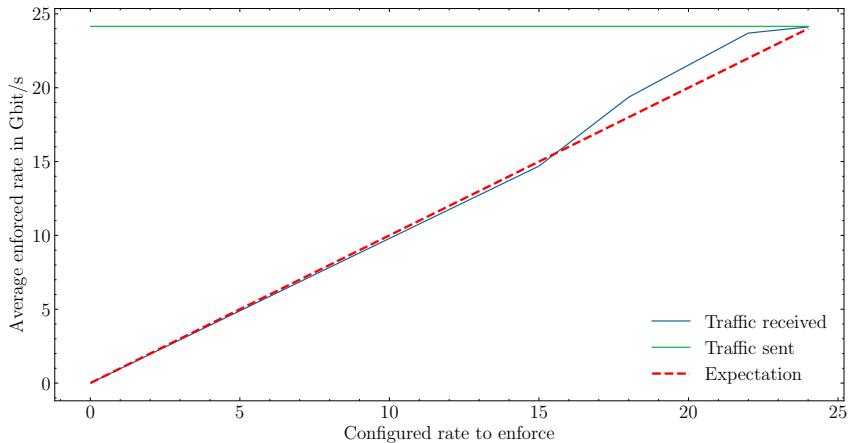
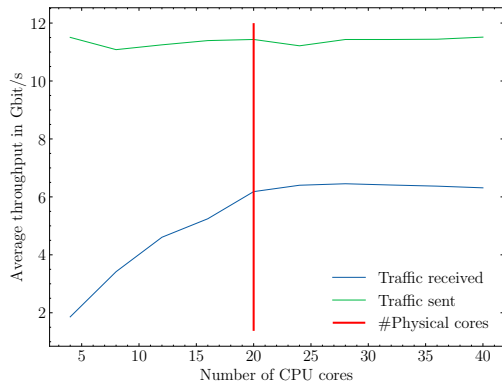
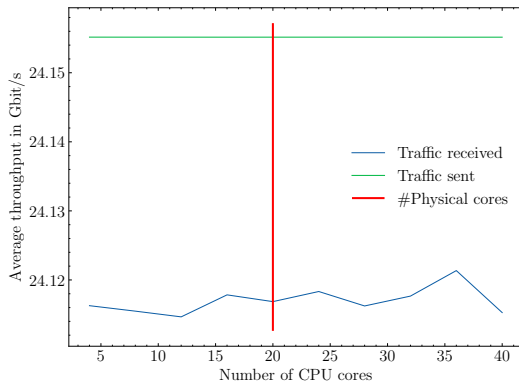


Figure 5: Rate enforcement of simple version

Real-World Experiments: Scaling Results



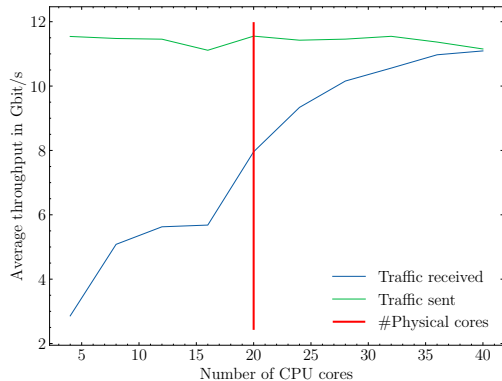
(a) 64-byte packets



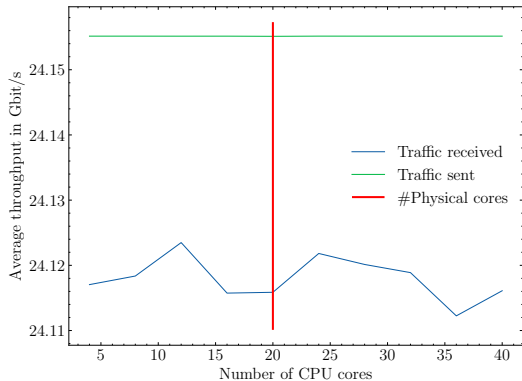
(b) MTU-sized packets (1514 bytes)

Figure 6: Throughput of simple version

Real-World Experiments: Scaling Results



(a) 64-byte packets



(b) MTU-sized packets (1514 bytes)

Figure 7: Baseline throughput: only forwarding packets

Future Work

- in-depth testing of high rates (≥ 20 Gbit/s)
- improve robustness of sharded version
- optimize for many queues with sparse rate distribution

BC-PQP in eBPF¹

Results:

- fully functional BC-PQP implementation
- multi-core support
- two synchronization approaches
- easy deployment and portability

Future work:

- in-depth testing of high rates
- improve robustness of sharded version
- optimize for many queues

¹<https://github.com/ftschirpke/BC-PQP-eBPF>

- [1] A. Tahir, P. Goyal, I. Marinos, M. Evans, and R. Mittal, **“Efficient Policy-Rich Rate Enforcement with Phantom Queues”**, in *Proceedings of the ACM SIGCOMM 2024 Conference*, Sydney NSW Australia: ACM, 2024, pp. 1000–1013.
- [2] S. Kunniyur and R. Srikant, **“An Adaptive Virtual Queue (AVQ) Algorithm for Active Queue Management”**, *IEEE/ACM Transactions on Networking*, vol. 12, no. 2, pp. 286–299, 2004.

Progress: TCP Results

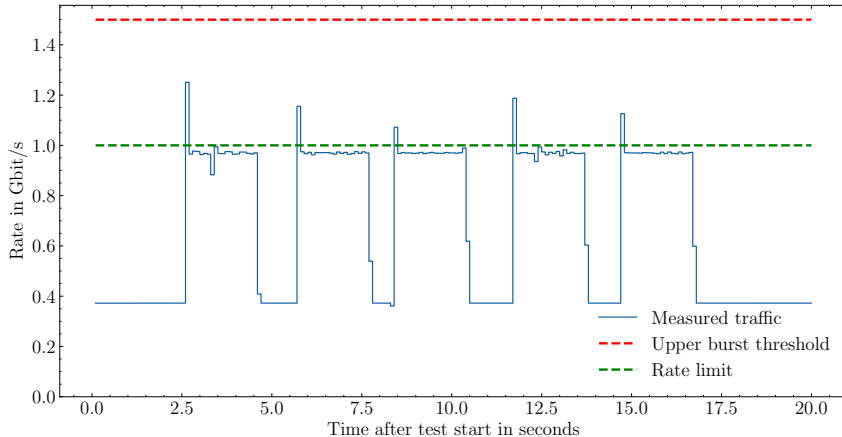


Figure 8: Throughput for bursty TCP traffic (RENO) - base sending rate of 400 Mbit/s with bursts of around 6 Gbit/s

Progress: Bursty Traffic Results

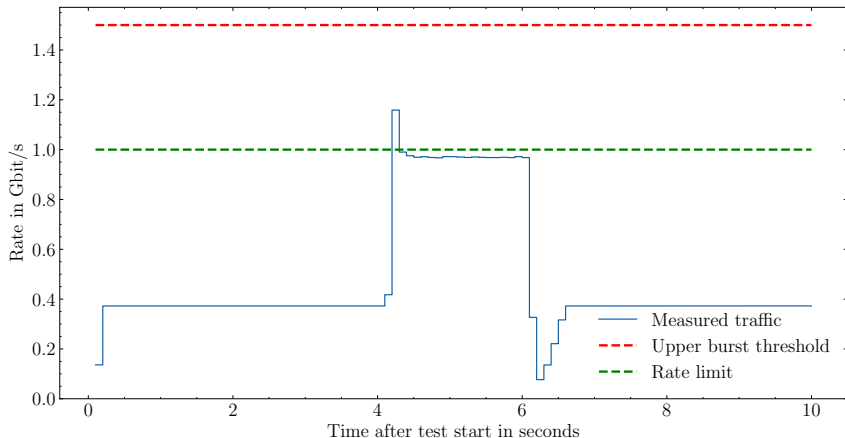


Figure 9: Sharded version: Throughput for bursty UDP traffic - base sending rate of 400 Mbit/s with bursts of around 6 Gbit/s

Real-World Experiments: Rate Enforcement Results

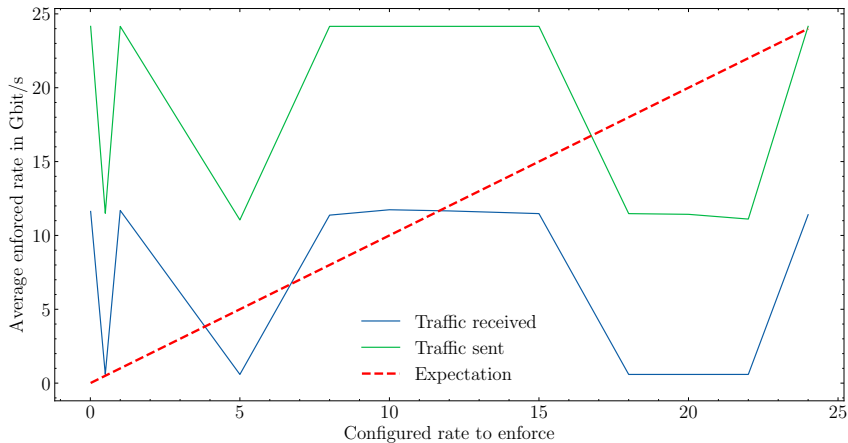
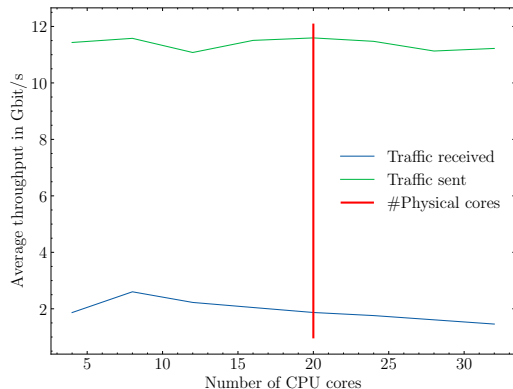
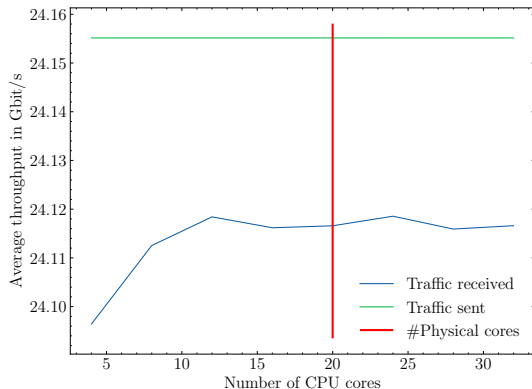


Figure 10: Rate enforcement of sharded version

Real-World Experiments: Scaling Results



(a) 64-byte packets



(b) MTU-sized packets (1514 bytes)

Figure 11: Throughput of sharded version